

AI Principles and Practice · Lecture 5

Unsupervised Learning and the Idea of a Representation

Week 5 · Phase I: Foundations and Ethics · LOs: LO1, LO2

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Where today goes

1. **No labels:** what the problem even is once the answer column is gone
2. **Clustering:** three algorithms, three different opinions about what a group is
3. **Choosing K:** a decision you defend, not a number the data hands you
4. **PCA:** your first machine that learns a new set of axes from the data itself
5. **Embeddings:** the single idea, geometry as meaning, that the rest of this course is built on

Same credit dataset as Weeks 3 and 4. This week we take away the labels **and** the group, then watch the geometry hand the group straight back.

What you'll be able to do by the end of today

These map to **LO1** (the principles and representations underpinning modern AI) and **LO2** (select, apply and critique core techniques). Read the verbs: not one of them is “recall”.

- **Contrast** K-means, hierarchical and GMM by the assumption each one smuggles in, not by their pseudocode.
- **Adjudicate** a disagreement between the elbow and the silhouette, and defend the K you settled on.
- **Interpret** a principal component in the language of the domain, rather than leaving it as “PC1”.
- **Critique** a learned representation for the structure it reproduces without being asked.
- **Construct** a 2D map of the credit applicants and argue for what its shape means.

We took the labels away. It sorted them anyway.

Hand a clustering algorithm the credit applicants from Week 3. No repayment label; that is the whole point of unsupervised learning. And, as always, no **group** column: we never let the model see the protected attribute.

Four features, standardised, nothing else. Ask only: where does this data naturally sit?

Project the result down to a picture and the two groups fall out into two regions of the plot. Nobody labelled them. Nobody named them. The geometry did it on its own, from the features we kept.

Nobody told it who these people were

We removed the label and the group. The structure the model found still tracks the group, because the group was never really gone: it was hiding in the correlations between the features we kept.

Today builds the two tools that surface this, clustering and PCA, and then asks the harder question: what is a representation, and whose fingerprints are on it?

Section 1

Learning without labels: proposing structure, not predicting it

What changes when the answer column disappears

Supervised learning had a target and, with it, a scoreboard: accuracy, a test set, a right answer to be wrong about. Take the label away and all of that goes with it.

Supervised: you are handed the answer and asked to reproduce it on data you have not seen. Success has a definition.

Unsupervised: you are handed only the data and asked to **propose** a structure. There is no answer key, so nothing can mark your proposal right.

The shift that trips everyone up

Clustering does not find the groups that are “really there”. It imposes a structure and reports how well the data tolerated it. The output is a hypothesis you now have to defend, not a fact you discovered.

K-means: a fast algorithm carrying strong opinions

The recipe is famously short: pick K centres, assign every point to its nearest one, move each centre to the mean of its members, repeat until nothing moves. It converges quickly and scales well.

That speed hides three assumptions you are agreeing to the moment you run it:

- clusters are **round** blobs of roughly equal spread,
- every point belongs **fully** to exactly one cluster,
- “near” means plain straight-line distance, so **scaling matters**.

The assumption is the model

K-means is not neutral machinery you point at data. It is a claim that your data is a handful of equal round balls. When that claim is wrong, the algorithm does not fail loudly; it just returns a confident, tidy answer that misreads the shape.

Standardise first, or distance lies to you

K-means measures distance, and distance has units. Leave income in pounds beside a debt ratio between 0 and 1, and income's raw magnitude drowns everything else out. The clustering becomes, in effect, a clustering on income alone.

Standardising each feature to zero mean and unit variance puts them on speaking terms. It is not a formality; it silently decides which features get a vote.

Evaluate before you trust

Whenever a clustering looks suspiciously like a sort on one column, check the scaling first. A “finding” that is really an artefact of unequal units is the most common way clustering embarrasses people.

Choosing K is choosing a claim about the data

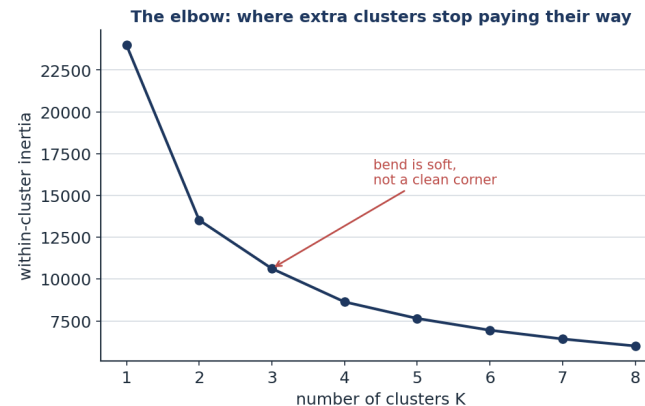
There is no K hiding in the data waiting to be read off. K is an argument you make, and the tools below inform that argument; they do not settle it.

The elbow: plot the within-cluster inertia as K grows. It always falls, because more centres always fit tighter. You look for the bend where each extra cluster stops earning its keep.

The silhouette: for each point, compare how close it sits to its own cluster against the nearest rival. Average it. High means cohesive and well separated; near zero means the borders are guesswork.

Two instruments, two different questions. When they agree, you have a case. When they disagree, you have a decision to make and a reason to give for it.

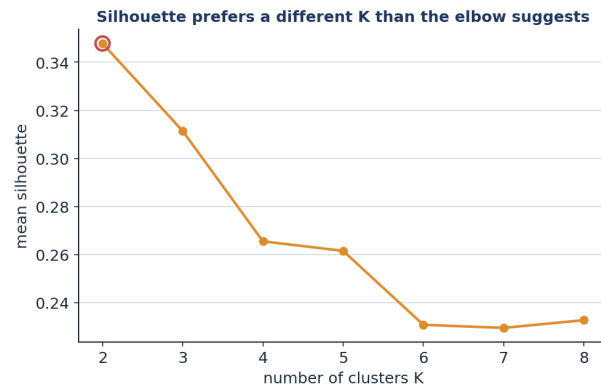
The elbow, on our own applicants



Inertia falls from 13,530 at $K=2$ to 6,000 by $K=8$. There is a gentle bend around three, but no clean corner. Real data rarely offers one.

The textbook elbow is a sharp right angle. This is what the method usually looks like in practice: a smooth curve down which you are invited to pick a corner that is not really there. The ambiguity is not a bug in your data; it is the honest state of most data.

The silhouette disagrees, and that is the lesson



Mean silhouette peaks at $K=2$ (0.35) and falls steadily after. Where the elbow hinted at three or four, the silhouette prefers two.

Two respected internal metrics, pointed at one dataset, return different answers. This is normal, and it is the point of showing you both.

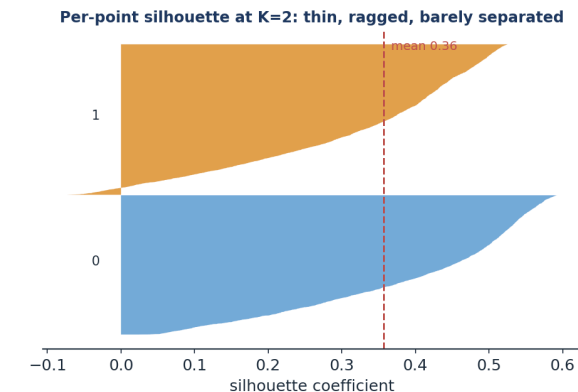
Note the **height**, too: 0.35 is not a strong silhouette. Values near one mean crisp, obvious clusters. Ours says the groups are real but soft, with plenty of points sitting on the fence.

So which K? Neither chart decides. You do, and you say why.

Read the silhouette per point, not just its average

The single averaged number hides the distribution. The knife plot opens it back up: one horizontal sliver per applicant, grouped by cluster, sorted by score.

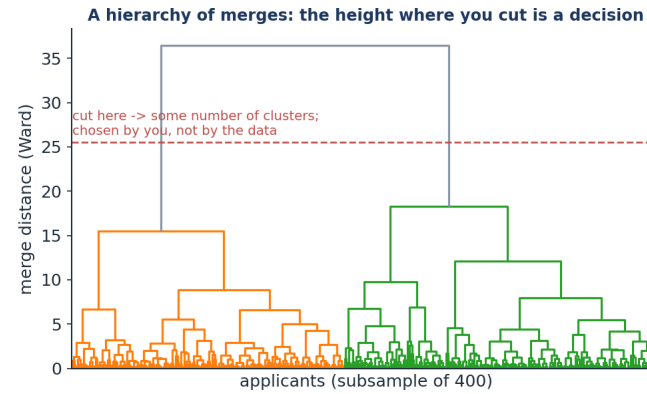
Wide, full blades mean a cohesive cluster. Thin, ragged ones, or bars that dip below zero, mean points that would sit more comfortably somewhere else.



At $K=2$ the blades are thin and uneven, with a long tail of near-zero points: exactly the fence-sitters the mean quietly averaged away.

A healthy mean can hide a sick cluster. Always look at the shape before you quote the number.

Hierarchical clustering: a tree, and a cut you own



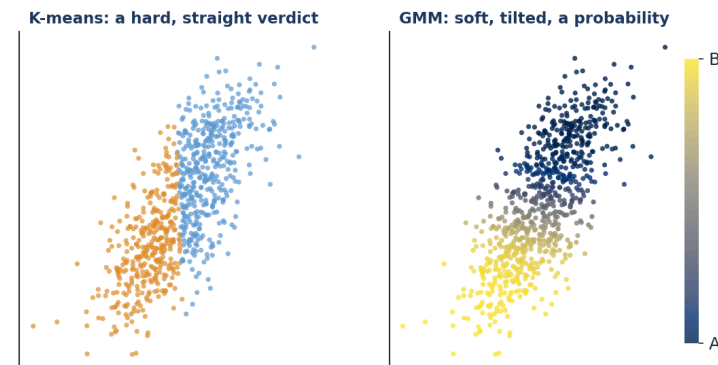
Applicants merged bottom-up by similarity (Ward linkage). The height of each join is how much cohesion that merge cost. Slice horizontally and you get clusters.

This method refuses to pick K for you and makes that refusal visible. It builds the whole hierarchy, from every-point-alone up to one-big-blob, and hands **you** the scissors. Where you cut is a modelling choice with your name on it, not an output of the algorithm.

GMM: from a hard verdict to a soft probability

K-means says: you are in cluster 2, full stop. A Gaussian mixture says: you are 78% cluster 2 and 22% cluster 1. It models each cluster as an ellipse with its own orientation and spread, and returns a **membership probability**.

K-means is the special case you get when you force those ellipses to be equal circles and round every probability to zero or one.



Two tilted, overlapping blobs. K-means slices them with a straight line and mislabels the overlap; GMM recovers the tilt and expresses the overlap as doubt.

The soft assignment is not a nicety; it is information. The applicants sitting at 50/50 are precisely the ones a human should look at, and a hard label would have hidden them.

Three algorithms, three theories of what a group is

Do not memorise these. Compare them by what each one assumes, and pick the one whose assumption matches your data.

	K-means	Hierarchical	GMM
Shape it expects	equal round balls	whatever linkage implies	tilted ellipses
Assignment	hard, one each	hard, set by the cut	soft probabilities
You must choose	K, up front	where to cut the tree	K, and covariance form
Where it misleads	non-round or unequal clusters	one bad early merge propagates	wrong K invents ellipses

Every one of these will return an answer on any data you give it, including data with no real clusters at all. The algorithm cannot tell you whether the structure exists; only whether the data tolerated the structure you imposed.

What cluster validity can, and cannot, tell you

Internal validity (silhouette, inertia) asks whether the clusters are tight and separated. It needs no ground truth, and it is blind to whether the clusters **mean** anything.

External validity compares your clusters to known labels. It is honest, but it needs labels you usually do not have; if you had them, you might not be clustering.

The trap in one line

A high silhouette certifies a tidy partition. It says nothing about whether that partition is useful, fair, or an artefact of your scaling. A clustering can be geometrically excellent and completely meaningless.

This is why the last word in unsupervised learning is never a metric. It is a human reading the clusters against the domain and judging whether they earn their keep.

Section 2

Dimensionality reduction and PCA: a representation you can watch being learned

Why high dimensions stop behaving

Our credit data has four features. Real project data has dozens or hundreds, and dimensions are not free.

As dimensions grow, points spread out until **everything** is roughly far from everything else. The gap between the nearest and the farthest neighbour shrinks, so “nearest” loses its bite, and every distance-based method, including the clustering we just did, quietly weakens.

The move this motivates

If most of the story lives in a few directions, find those directions and drop the rest. Compress first, then cluster or visualise in a space where distance means something again.

PCA: rotate the axes to where the data actually spreads

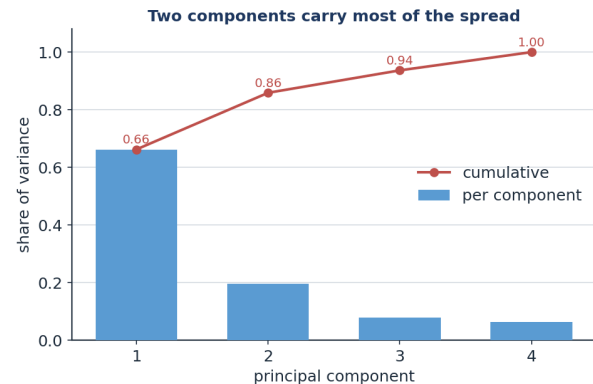
Your features came with axes chosen by whoever built the spreadsheet. PCA throws those away and asks a better question: along which direction does the data vary the most? Call that PC1. Then the next most, at right angles to it: PC2. And so on.

Keep the first few and you keep most of the variation in far fewer numbers. The axes are no longer income and debt; they are **combinations** the data itself picked out.

This is the key move

Nobody chose these new axes by hand. The algorithm learned them from the covariance of the data. That is what makes PCA your first **learned representation**, and why it belongs at the centre of this lecture rather than in a footnote about compression.

How much are you willing to throw away?



PC1 carries 66% of the variance, PC2 another 20%. Two axes hold 86% of everything; the last two carry the remaining scraps.

The scree plot is the honest accountant of PCA. Each bar is the share of variance one component explains; the line is the running total.

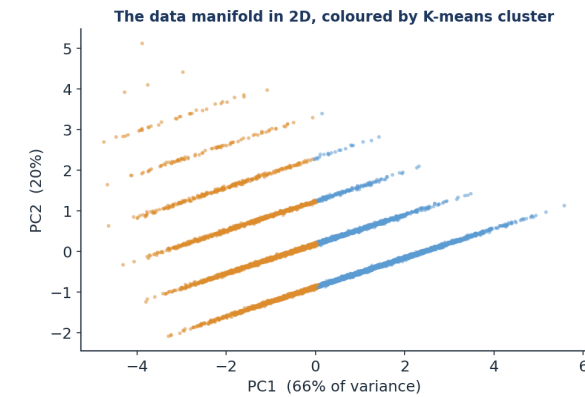
Reducing four features to two here costs you 14% of the variance and buys you a picture you can actually look at. Whether that trade is worth it is a judgement about **your** problem, not a rule.

Note the assumption you have just accepted: that variance is where the signal lives. Often true. Not always.

The manifold, finally visible in 2D

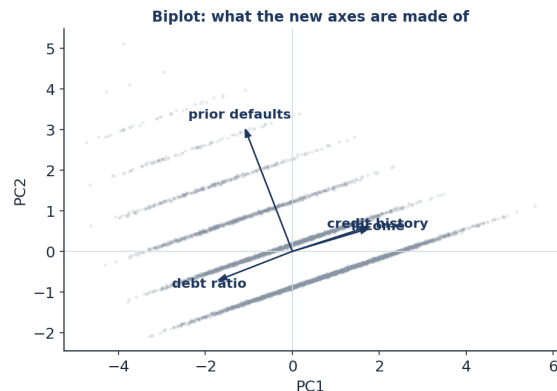
Four dimensions cannot be drawn. Two can.
Projecting onto PC1 and PC2 turns the applicant pool into something you can see, and structure that was invisible in the raw table appears as regions, gradients and gaps.

Coloured by K-means cluster, the split runs cleanly along PC1: the algorithm and the projection agree on the main axis of variation. That agreement is reassuring, and, in a moment, revealing.



The credit applicants on their two learned axes, coloured by cluster. The primary division lies along PC1, left to right.

Interpret the axes, or the picture is decoration



Loadings for PC1: income +0.55 and credit history +0.55 pull one way; debt ratio -0.53 and prior defaults -0.34 pull the other.

A component is a weighted blend of the original features, and the biplot shows the recipe. Read PC1 here and it is unmistakable: high income and long history at one end, high debt and past defaults at the other.

That is not “PC1”. That is a **financial-health axis** the data drew for itself. Naming it in the language of the domain is the difference between a representation you understand and a plot you merely printed.

The higher-order skill this week is exactly this translation: from a vector of loadings to a sentence a domain expert would nod at.

PCA learned a representation; say what that means

Step back from the mechanics. PCA took raw features and returned a new coordinate system, fitted to the data, in which each point is now described by a short vector of numbers.

The applicant did not change. Their **description** did: from four measured quantities to two coordinates on axes the algorithm invented. That new description is the representation.

Everything downstream, the clustering, the visualisation, any model you build next, now runs on this representation rather than on the raw data. The choice of representation quietly shapes all of it.

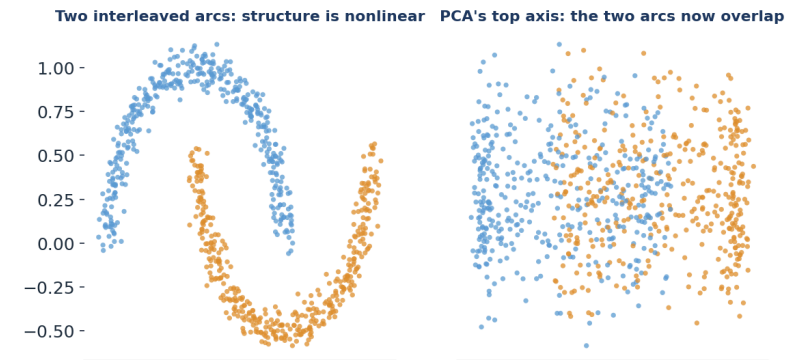
The sentence to carry out of the room

A representation is a learned answer to “how should this data be described?”. Change the representation and you change what every later step can even see.

Where a straight axis destroys the structure

PCA is linear: its axes are straight lines. When the real structure curves, a straight projection can smear it into nonsense.

The two interleaved arcs on the left are cleanly separable, but not by any straight line. Flatten them onto PCA's top axis and the two classes land on top of each other. The variance was real; it just was not the **signal**.



Nonlinear structure that PCA cannot preserve: projecting the arcs onto one straight axis makes them overlap.

Evaluate PCA by its assumption, not its popularity. When it fails, it fails quietly, returning a confident, low-dimensional picture that has thrown the real structure away. This is exactly the gap that nonlinear methods, and learned embeddings, exist to close.

Section 3

Embeddings: the idea that unifies the whole course

From components to embeddings: one idea, generalised

PCA gave every applicant a short vector on learned axes. Hold that thought and drop the requirement that the axes be straight, or few, or human-named. What remains is the whole idea of an **embedding**.

An embedding places each item, an applicant, a word, an image, a molecule, as a point in a space that some method learned, so that position carries meaning. PCA is the simplest honest example. Everything more powerful later in the course is a richer version of the same move.

Geometry as meaning: distance and direction do the work

Once data are points in a learned space, the geometry of that space **is** the model of similarity.

Distance becomes similarity: points near each other are things the representation considers alike. Nearest-neighbour search, recommendation and retrieval all run on this one fact.

Direction can become meaning: in good embeddings, consistent offsets encode consistent relationships, which is why “king minus man plus woman” famously lands near “queen”.

This is the sentence the whole module turns on: a representation is a space you have arranged so that the questions you care about become geometry. Get the space right and the hard problem becomes measuring a distance.

The representation is the product

It is tempting to think the model is the clever part and the representation is preparation. Later weeks invert that. Modern systems spend most of their effort **learning a good representation**; the task-specific layer on top is often small.

A model almost never reasons over raw pixels or raw characters. It reasons over an embedding, and the quality of that embedding sets the ceiling on everything above it.

Why this lecture sits where it does

Weeks 2 to 4 built and interrogated models. This week names the thing those models were quietly standing on the whole time. From here on, “what representation is this running on?” is a question you ask by reflex.

Linear now, nonlinear soon: trade-offs to weigh

PCA is the linear anchor. The methods you meet later trade its guarantees for reach, and each trade costs something worth knowing now.

	What it buys	What it costs
PCA (linear)	fast, interpretable axes, faithful global structure	misses any curved or nonlinear structure
t-SNE / UMAP	beautiful local neighbourhoods, clusters pop out	distances between clusters can be meaningless
Learned embeddings	task-tuned, richly expressive	opaque axes, needs data and compute, harder to audit

A cardinal error with t-SNE and UMAP: reading the gaps between clusters as real distances. They optimise local neighbourhoods and will happily stretch or squash the space between groups. Critique the method before you narrate its picture.

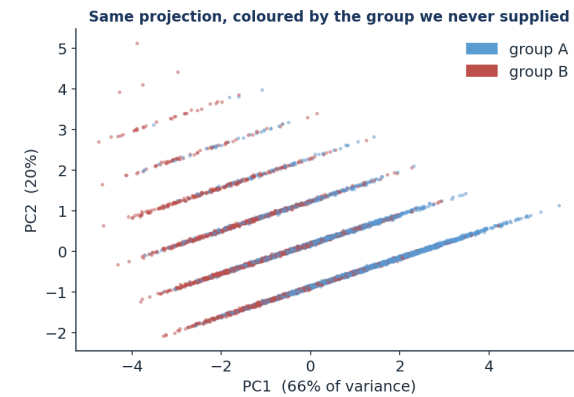
Section 4

Whose fingerprints are on the representation?

The reveal: recolour the same plot by group

Here is the picture from earlier, unchanged: the same applicants, the same PC1 and PC2, the same learned axes. Only the colouring is different. Now each point is tinted by the protected group we never supplied.

The groups do not smear together. They separate, largely along PC1: the very financial-health axis the data drew for itself.



The identical projection, coloured by group. Group B sits low and left, group A high and right, along the main axis of variation.

Put a number on it: the representation leaks the group

A picture invites argument, so measure it. Fit a tiny model that sees **only** PC1 and tries to name each applicant's group.

Quantity	Value
Group recoverable from PC1 alone	0.71
Base rate (guess the majority group)	0.43
Recall: Week 4, group recoverable from the supervised score	0.70

Read the last two rows together. An unsupervised representation, built with no labels and no group, encodes the protected attribute at almost exactly the rate last week's supervised model did. Removing the column removed the **word**, not the information.

Why it happens: proxy leakage, now in geometric form

This is the same mechanism from Weeks 3 and 4, wearing new clothes. Group left visible tracks in income, in history length, in debt, because the world that generated the data is unequal.

PCA does not seek out the group. It seeks out variance. But the largest axis of variance in this data **is** the axis along which the groups differ, so the representation reconstructs the divide as a matter of pure arithmetic.

Analyse the chain

Unequal world, correlated features, variance aligned with the inequality, an axis that recovers the group. No malice, no bug, no protected column. Just correlation, faithfully compressed.

A representation is never neutral

It is easy to treat PCA as objective: no labels, no choices, just the data speaking. That framing is exactly the danger. The representation inherits **every** correlation in the data, including the ones you spent Week 3 trying to design out.

So the fairness question does not end when you drop the sensitive attribute. It reopens the moment you learn a representation, because the representation can rebuild the attribute from what is left, and then quietly pass it to everything downstream.

Whatever a dataset's protected attribute is, assume the representation can approximate it, and check. "We removed that field" is a statement about the schema, not about the geometry.

The limits of unsupervised evaluation, revisited

Bring back the silhouette from the first half, and notice what it could never have told you.

Our clustering scored a perfectly respectable silhouette. That number certified geometric tidiness and nothing else. It was structurally incapable of noticing that the tidy split it rewarded was a split along a protected group.

The evaluative limit

Internal metrics measure shape, not justice. A clustering can be cohesive, well separated, high-scoring, and discriminatory all at once, and no silhouette will ever flag the last of those.

Which returns us to the same verdict as the clustering half, now with teeth: the last word is a human, reading the representation against the harm it could do.

Higher-order prompt

Take the credit applicants we have been clustering. Before the lab, write down two sentences.

1. The structure you **hope** clustering and PCA will find, the useful segments, the interpretable axes.
2. The structure you would be **uncomfortable acting on** if it fell out instead: the division that quietly tracks the protected group, or something standing in for it, that you did not intend to model.

If you cannot name the second sentence, you are not ready to trust the first. The whole point of today is that a method with no labels and no bad intent can still hand you the second one.

Section 5

The lab: finding structure, and interrogating it

Today's lab, end to end

You take **the credit applicants** and do four things, in order. The deliverable is an interpretation, not a gallery of plots.

1. **Cluster:** run K-means; be honest about scaling and about what a cluster here would mean.
2. **Validate:** compute the elbow and the silhouette, watch them disagree, and choose a K you can defend.
3. **Reduce:** apply PCA, read the scree, project to 2D, and name PC1 and PC2 in domain terms.
4. **Interrogate:** recolour your 2D map by any sensitive attribute you have, and check whether your representation leaks it.

Cluster and validate

cluster.py

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

Xs = StandardScaler().fit_transform(X) # scale FIRST

for k in range(2, 9):
    km = KMeans(k, n_init=10, random_state=7).fit(Xs)
    print(k, km.inertia_,
          silhouette_score(Xs, km.labels_))
```

K, inertia, silhouette

```
2 13530.2 0.348
3 10629.9 0.311
4 8637.9 0.265
... elbow soft,
    silhouette says 2
```

Do not automate the choice of K. Print both columns, look at them, and write one sentence defending the K you pick.

Reduce, project, and name the axes

reduce.py

```
from sklearn.decomposition import PCA

pca = PCA().fit(Xs)
print(pca.explained_variance_ratio_)
# -> [0.66, 0.20, 0.08, 0.06]

Z = PCA(n_components=2).fit_transform(Xs)
# Z is your learned 2D representation

# read the recipe for PC1:
for name, w in zip(features, pca.components_[0]):
    print(name, round(w, 2))
```

PC1 loadings

income	+0.55
credit history	+0.55
debt ratio	-0.53
prior defaults	-0.34

-> a "financial
health" axis

A plot with axes called PC1 and PC2 is unfinished. Name them, or you have not interpreted anything.

Interrogate: does your map leak what it should not?

leak.py

```
from sklearn.linear_model import LogisticRegression

pc1 = Z[:, [0]] # the top axis alone
probe = LogisticRegression().fit(pc1, sensitive)
print("recoverable:", probe.score(pc1, sensitive))
# base rate = sensitive.mean()

# then LOOK: recolour the 2D scatter
plt.scatter(Z[:,0], Z[:,1], c=sensitive, s=8)
```

The actual exercise

Not “make a 2D plot”. Rather: make it, then attack it. If a one-feature probe can recover a protected attribute from your top component, your representation is a proxy, and you say so in the report.

A strong interpretation versus a box-tick

Box-tick

- “We found 3 clusters.”
- “PCA explained 86% of variance.”
- Prints a scatter plot, PC1 vs PC2.
- Ends there.

Strong

- Names what each cluster **is** in the domain, and which ones are shaky.
- Reads PC1 as a financial-health axis, with evidence from the loadings.
- Reports that PC1 recovers the group at 0.71, and treats that as a finding.
- States which structure it would **refuse** to act on, and why.

The plot is the easy 20%. The grade lives in the interpretation: what the structure means, what it hides, and what you will not do with it.

What you're handing in, and where it goes next

In the lab

- A defended choice of K , with the elbow and silhouette shown.
- A 2D PCA map with **named** axes.
- A leakage check on any sensitive attribute, with the number stated plainly.

Where it goes next

- The same four moves return as **Elective 2** of your assignment, assessed there on the dataset the team assigns you.
- The leakage result feeds straight back into the Week 3 fairness work and the Week 4 governance note.

The credit data is the rehearsal: run these moves here until they are automatic. The marks are for the argument on top, what the structure means, what it hides, and what you would refuse to do with it.

Five things worth leaving with

1. **Unsupervised learning proposes structure; it does not discover truth.** Clustering imposes a shape and reports how well the data tolerated it. The output is a hypothesis you must defend, not a fact.
2. **Choosing K, or where to cut, or which method, is a modelling decision with your name on it.** The elbow and silhouette inform the argument; when they disagree, you decide and you justify.
3. **PCA is your first learned representation, not just compression.** It invents a new coordinate system from the data, and everything downstream then runs on that system rather than on the raw features.
4. **Geometry as meaning is the idea the whole course rests on.** Embeddings turn similarity into distance and relationships into direction; get the space right and the hard problem becomes measuring a distance.
5. **A representation is never neutral.** It inherits every correlation in the data, so it can rebuild a protected attribute you dropped, and no internal metric will ever warn you. The last word is a human, not a silhouette.

Where to read next

Papers and primers

- Jolliffe & Cadima (2016), **Principal Component Analysis: a Review and Recent Developments**: the clearest modern account of what PCA is and is not.
- Rousseeuw (1987), **Silhouettes**: the original silhouette paper, short and readable.
- van der Maaten & Hinton (2008), **Visualizing Data using t-SNE**: the method, and its own honest caveats.
- McInnes, Healy & Melville (2018), **UMAP**: the newer workhorse; read the “distances between clusters” warnings.
- Mikolov et al. (2013), **Distributed Representations of Words and Phrases**: where “geometry as meaning” became concrete.

Books and further reading

- James, Witten, Hastie & Tibshirani, **An Introduction to Statistical Learning**: chapters 10 (unsupervised) and 12; free PDF, worked in R and Python.
- Bishop, **Pattern Recognition and Machine Learning**: chapter 9 for GMMs and the EM view of clustering.
- Géron, **Hands-On Machine Learning** (3rd ed.): chapters 8 to 9 for practical scikit-learn clustering and PCA.
- Wattenberg, Viégas & Johnson (2016), **How to Use t-SNE Effectively** (distill.pub): interactive, and a fine antidote to over-reading embeddings.

Read Jolliffe & Cadima before the lab. Knowing exactly what PCA optimises is what lets you say, with a straight face, why your representation leaked the group.